

Data Analysis and Reporting Using Power BI

Executive Summary

This project builds a complete synthetic university analytics dataset and a Power BI-ready reporting package covering enrollment, academic performance, attendance, finance, faculty workload, and department budgets.

Dataset Description

The cleaned dataset contains 1000 students, 70 courses, 50 faculty members, 12500 enrollment records, 22577 attendance records, and 5994 payment records across 2023-2026.

Data Model Design

The model uses dimensions for Students, StudentProfile, Departments, Courses, Faculty, and Semesters, facts for Attendance, Payments, and DepartmentBudget, and Enrollment as the bridge/fact table resolving the conceptual many-to-many relationship between Students and Courses.

Power Query Transformations

The raw data intentionally includes inconsistent capitalization, blanks, duplicate records, invalid marks, date text, and payment-status inconsistencies. Cleaning standardizes text, removes duplicates, corrects types, validates ranges, and derives risk categories.

DAX Measures

The measure library contains 40 documented measures stored in `\_Measures`, including KPIs, rates, growth calculations, ranking, dynamic titles, TOPN, CONCATENATEX, and USERELATIONSHIP examples.

Dashboard Design

The report specification includes six required pages plus an optional drill-through page with slicers, bookmarks, dynamic titles, conditional formatting, report-page tooltips, and cross-filtering.

Main Findings

Highest enrollment department: English has the largest student count. Recommended action: Review section sizes and faculty allocation.

Highest failure course: C018 has the highest failure rate. Recommended action: Add tutoring and review course design.

Attendance relationship: Lower attendance bands show lower average final marks. Recommended action: Trigger advising below 75% attendance.

Outstanding payments: Outstanding balance totals 125,733,250. Recommended action: Segment outreach by payment risk level.

Budget utilization: D007 has the highest utilization in 2023. Recommended action: Review mid-year budget reallocations.

Scholarship distribution: 100.0% of students receive a scholarship. Recommended action: Track outcomes by scholarship type.

Faculty workload: Enrollment links faculty to taught courses for workload analysis. Recommended action: Monitor distinct courses and enrollments per faculty.

Graduation eligibility: Credit and GPA measures identify students approaching eligibility.

Recommended action: Add drill-through student review page.

Payment status: Paid, partial, and unpaid categories provide finance risk visibility. Recommended action: Use respectful outreach workflows.

Semester trend: Semester fields support enrollment and performance trend pages. Recommended action: Monitor semester-over-semester growth.

Limitations

Power BI Desktop was not installed in the execution environment, so the `.pbix`, live Power BI Service URL, screenshots from the actual dashboard, and tenant publication checks remain manual completion steps.

Conclusion

The package is ready for Power BI Desktop authoring using the cleaned workbook, relationship documentation, DAX library, Power Query code, and report-page build guide.